

Dimension Reduction

Faculty of Information Technology, Monash University, Australia

FIT5149 week 7

- 1 The Curse of Dimensionality
- 2 Principal Component Analysis
- 3 T-SNE
- 4 Principal Component Regression

- Weekly learning outcomes:
 - ▶ Differentiate between supervised learning and unsupervised learning.
 - ▶ Identify the issues of high dimensions in analysis.
 - ▶ Understand the principle of PCA, PCR, and t-SNE.
 - ▶ Apply PCA and t-SNE to summarizing a large set of correlated variables with a small number of representative variables.
 - ▶ Interpret the loading and scoring matrices.

High-Dimensional Data

- High-dimensional: more features than observations, i.e., $p \gg n$
 - ▶ Genetic data: If we consider genes as variable, this means that our observations are in a space with thousands of dimensions.
 - Predict blood pressure with *single nucleotide polymorphisms* (SNPs)

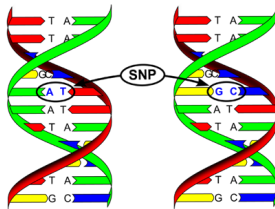
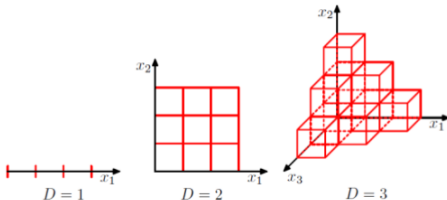
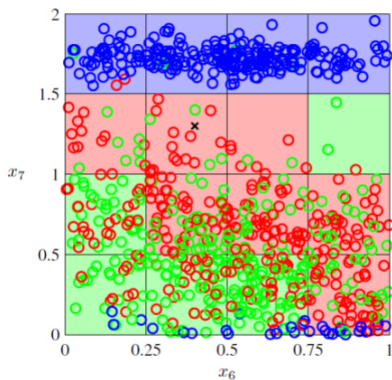


Figure: The picture is from <http://www.viagenefertility.com/photos/snp.png>

- [illegible]

Figure: The picture was generated by wordclouds.com

The Curse of Dimensionality: Complexity Increases w.r.t. #Features.¹



The number of regions increases exponentially w.r.t. the number of features.

¹The figures used from this slide are from the textbook Machine Learning and Pattern Recognition.

The Curse of Dimensionality: Risk of Overfitting

- When p is as large as, or larger than, n , least squares won't work: it yields coefficients that fit the training data perfectly, with zero residuals.

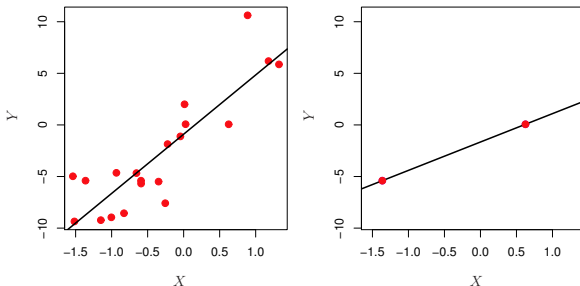


Figure: Left: Least squares regression in a low-dimensional setting. Right: Least squares regression with $n = 2$ observations.

- Problem: when $p > n$ or $p \approx n$, least squares regression is too flexible and overfits the data.

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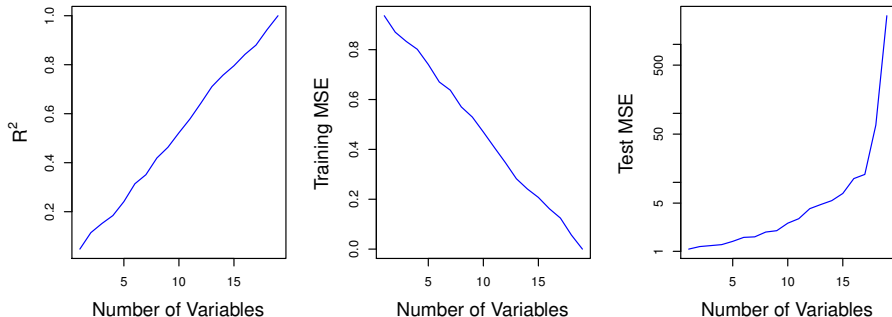
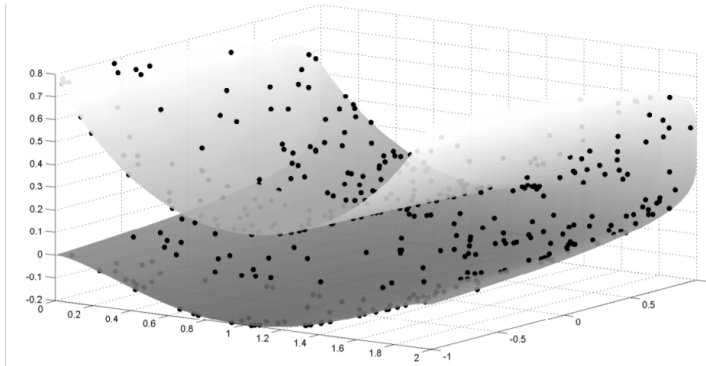


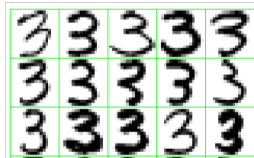
Figure: Simulated example with $n = 20$ training observations: as features unrelated to the outcome are added, training R^2 rises and training MSE falls, but test MSE grows.

- Problem: when $p > n$ or $p \approx n$, least squares regression is too flexible and overfits the data.

The Curse of Dimensionality: Manifold Assumption²



(a)



(b)

$$\begin{aligned}
 \hat{f}(\lambda) &= \bar{x} + \lambda_1 v_1 + \lambda_2 v_2 \\
 &= \boxed{3} + \lambda_1 \cdot \boxed{3} + \lambda_2 \cdot \boxed{3}
 \end{aligned}$$

(c)

²Figures used in this slides are from https://www.ml.uni-saarland.de/code/GraphDemo/HeinLuxburg_SlidesSemisupervisedLearning.pdf

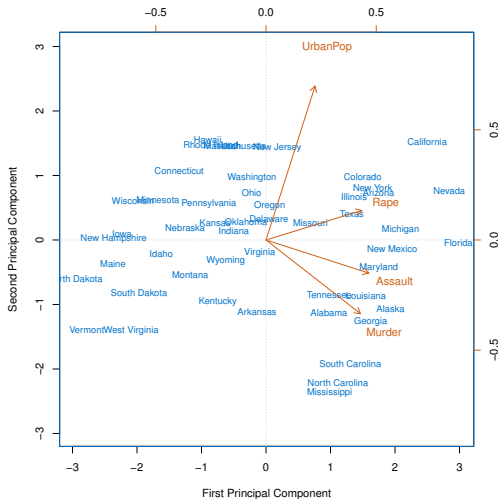
Outline



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- 1 The Curse of Dimensionality
- 2 Principal Component Analysis**
- 3 T-SNE
- 4 Principal Component Regression

Why Dimension Reduction?



- Visualisation.
- Preprocessing.
- Overcome the curse of dimensionality.

PCA: Principal Component Analysis³

- PCA produces a low-dimensional representation of a dataset.
 - ▶ A sequence of linear combinations of the variables that have maximal variance and are mutually uncorrelated.
 - ▶ The first principal component is that (normalised) linear combination of the variables $(X_1, X_2, \dots, X_p)$ with the largest variance.

$$Z_1 = \phi_{1,1}X_1 + \phi_{2,1}X_2 + \dots + \phi_{p,1}X_p$$

- ▶ We refer to the elements $\phi_{1,1}, \dots, \phi_{p,1}$ as the loadings of the first principal component, and Z_i is referred to as the principal component scores of $\mathbf{X}$ on the first principal component.

³[https:](https://scholarscompass.vcu.edu/cgi/viewcontent.cgi?article=1006&context=ssor_pubs)

[//scholarscompass.vcu.edu/cgi/viewcontent.cgi?article=1006&context=ssor_pubs](https://scholarscompass.vcu.edu/cgi/viewcontent.cgi?article=1006&context=ssor_pubs)

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PCA: Compute the First Principal Component

Suppose we have a $n \times p$ data set X . Since we are only interested in variance, we assume that each of the variables in X has been centred to have mean zero (that is, the column means of X are zero).

$$z_{i,1} = \phi_{1,1}x_{i,1} + \phi_{2,1}x_{i,2} + \cdots + \phi_{p,1}x_{i,p}$$

- $z_{i,j}$ has mean zero since each of the $x_{i,j}$ has mean zero, due to the scaling.
- The sample variance of the $z_{i,1}$ can be written as

$$\frac{1}{n} \sum_{i=1}^n (z_{i,1} - 0)^2$$

PCA: Compute the First Principal Component

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$$z_{i,1} = \phi_{1,1}x_{i,1} + \phi_{2,1}x_{i,2} + \cdots + \phi_{p,1}x_{i,p}$$

- The optimisation problem

$$\underset{\phi_1}{\text{maximise}} \left\{ \frac{1}{n} \sum_{i=1}^n (\phi_1^T \mathbf{x}_i - 0)^2 \right\} \quad (1)$$

$$\text{s.t.} \sum_{j=1}^p \phi_{j,1}^2 = 1 \quad (2)$$

- Why $\sum_{j=1}^p \phi_{j,1}^2 = 1$?

PCA: Compute the First Principal Component

The optimisation problem has the following generic form, which does not require to make data centred to have zero means.

- The (generic) optimisation problem:

$$\underset{\boldsymbol{\phi}_1}{\text{maximise}} \left\{ \frac{1}{n} \sum_{i=1}^n (\boldsymbol{\phi}_1^T \mathbf{x}_i - \boldsymbol{\phi}_1^T \bar{\mathbf{x}})^2 \right\} \quad (3)$$

$$\text{s.t.} \sum_{j=1}^p \phi_{j,1}^2 = 1 \quad (4)$$

- Solution: $\mathbf{S}\boldsymbol{\Phi}_1 = \lambda_1 \boldsymbol{\Phi}_1$, where $\mathbf{S} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$ is the covariance matrix of the data, $\lambda_1 \in \mathbb{R}$ denotes an eigenvalue and $\boldsymbol{\Phi}_1$ denotes the eigenvector of λ_1 . The problem is reduced to find the eigenvector of the largest eigenvalue λ_1 .
- Optimisation method: Singular-value Decomposition (SVD) of the matrix S , a standard techniques in linear algebra, see http://web.mit.edu/be.400/www/SVD/Singular_Value_Decomposition.htm, or the power method.

PCA: Visualise First Principal Component

- The first principal component is that (normalised) linear combination of the variables with the largest variance.

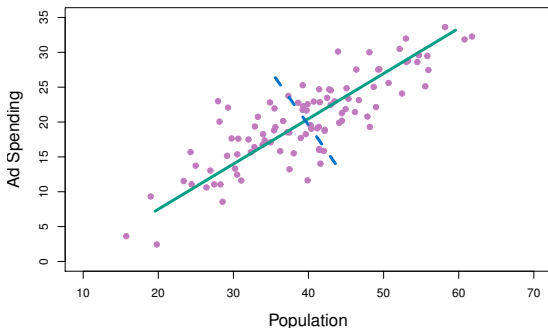


Figure: The population size (pop) and ad spending (ad) for 100 different cities are shown as purple circles. The green solid line indicates the first principal component, and the blue dashed line indicates the second principal component.

PCA: Visualise the First Principal Component

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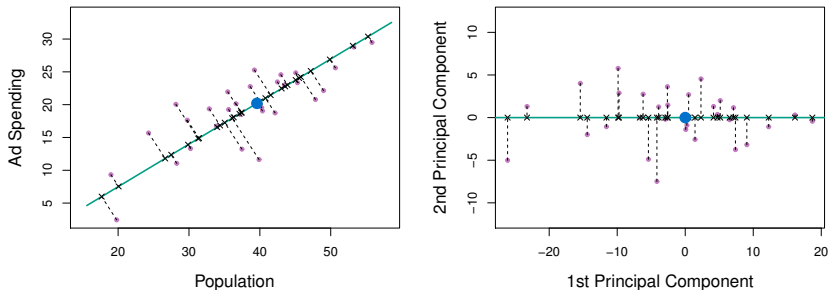
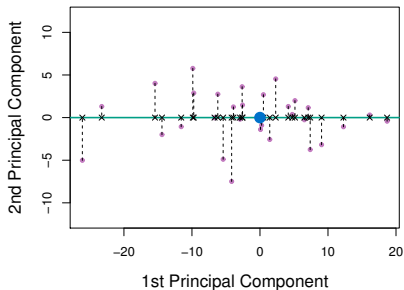
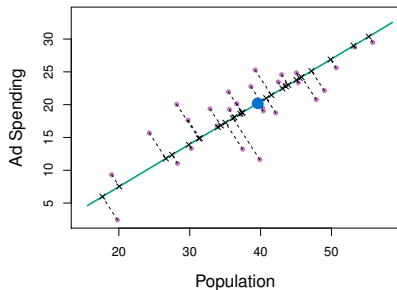


Figure: A subset of the advertising data. **Left:** The first principal component, with projected coordinates for each point, is shown in green. These distances are represented using the black dashed line segments. **Right:** The left-hand panel has been rotated so that the first principal component lies on the x-axis.

PCA: Visualise the First Principal Component

- The first principal component is that (normalised) linear combination of the variables with the largest variance.



- Given the loadings for the first principal component, we can compute the principal component scores.

$$z_{i,1} = 0.839 \times (pop_i - \overline{pop}) + 0.544 \times (ad_i - \overline{ad})$$

PCA: Visualise the First Principal Component

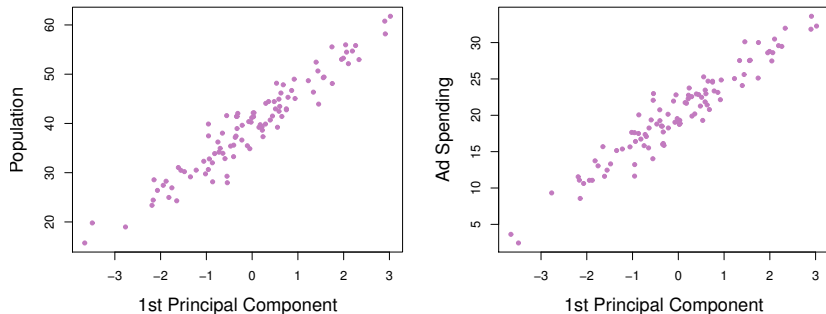


Figure: Plots of the first principal component scores $z_{i,1}$ versus pop and ad. The relationships are strong.

- **pop** and **ad** have approximately linear relationship
- The above plots show a strong relationship between the first principal component and the two features.

PCA: the Second Principal Component

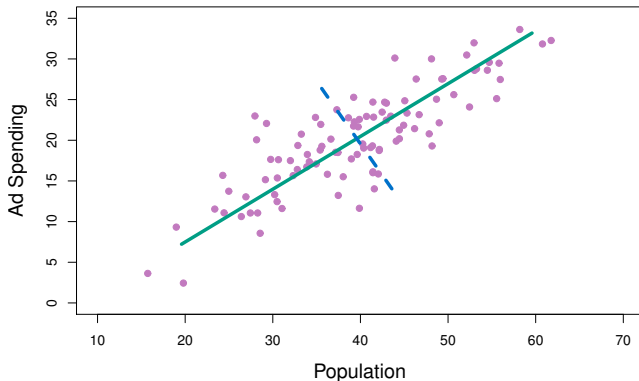
- The second principal component is the linear combination of $X_1, X_2, \dots, X_p$ that has maximal variance among all linear combinations that are uncorrelated with Z_1 .

$$Z_{i,2} = \phi_{1,2}X_{i,1} + \phi_{2,2}X_{i,2} + \dots + \phi_{p,2}X_{i,p}$$

where ϕ_2 is the second principal component loading vector.

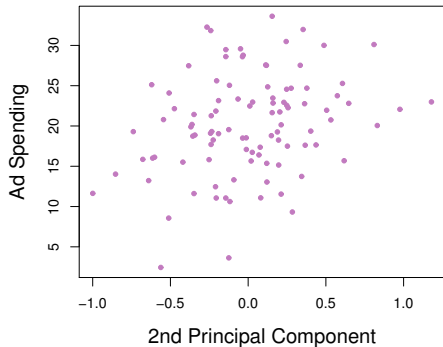
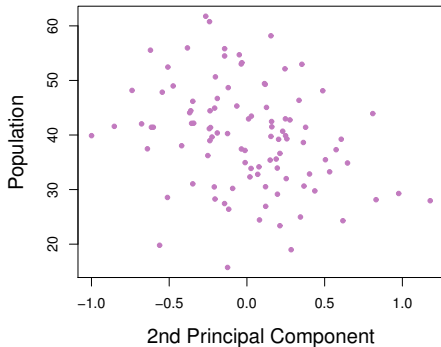
- Constraining Z_2 to be uncorrelated with Z_1 is equivalent to constraining the direction ϕ_2 to be orthogonal (perpendicular) to the direction ϕ_1 , namely $\phi_2^T \phi_1 = 0$. You will get orthogonal eigenvectors directly from SVD, but need the constraint for the other optimisation methods.
- The principal component directions $\phi_1, \phi_2, \phi_3, \dots$ are the ordered sequence of eigenvectors of the matrix X . There are at most $\min(n-1, p)$ principal components.

PCA: the Second Principal Component



- The first two components contain all the information that is in **pop** and **ad**.
- The first component contains the most information.
- The second component scores are much closer to zero → capture less information

PCA: the Second Principal Component



- Little relationship between the second principal component and these two predictors.

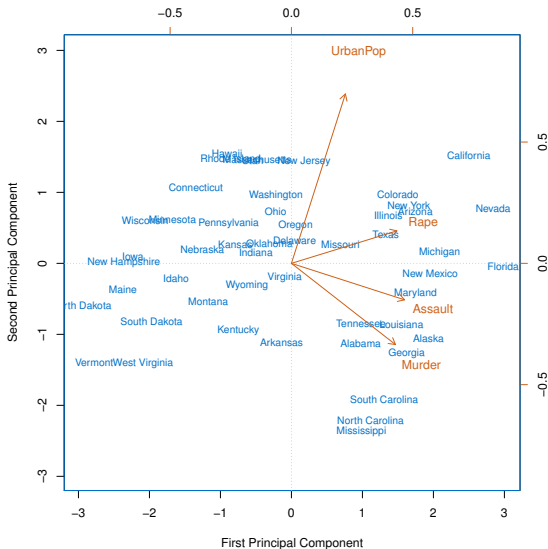
PCA: Illustration — Data

- USArrests data: For each of the fifty states in the United States, the data set contains the number of arrests per 100, 000 residents for each of three crimes: **Assault**, **Murder**, and **Rape**. We also record UrbanPop (the percent of the population in each state living in urban areas).
- The principal component score vectors have length $n = 50$, and the principal component loading vectors have length $p = 4$.

	PC1	PC2
Murder	0.5358995	-0.4181809
Assault	0.5831836	-0.1879856
UrbanPop	0.2781909	0.8728062
Rape	0.5434321	0.1673186

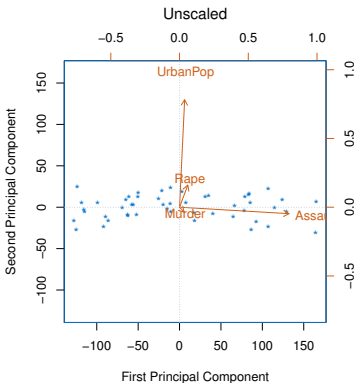
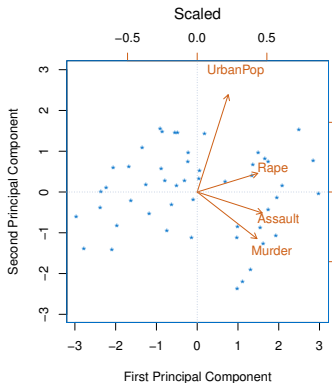
- PCA was performed after standardising each variable to have mean zero and standard deviation one.

PCA: Illustration — biplot



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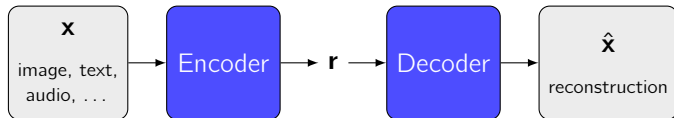
Scaling of the Variables



- The four variables (Murder, Rape, Assault and UrbanPop) have variance: 18.97, 87.73, 6945.16, and 209.5.
- If the variables are in different units, scaling each to have standard deviation equal to one is recommended.

Autoencoder⁴

- An autoencoder compresses an input $\mathbf{x}$ into a code $\mathbf{r}$ (encoder) and reconstructs $\hat{\mathbf{x}}$ from $\mathbf{r}$ (decoder).
- No labels are needed: the target is the input itself, so it is unsupervised.

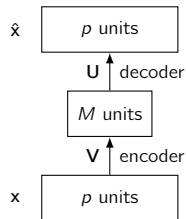


- Training objective: minimise the reconstruction error $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$ over the training set.
- If $\mathbf{r}$ has fewer dimensions than $\mathbf{x}$, the bottleneck forces the encoder to keep only the most informative directions: this is dimension reduction.

⁴<https://lars-quaedvlieg.github.io/blog/2024/cs330-stanford-upt-rbm>

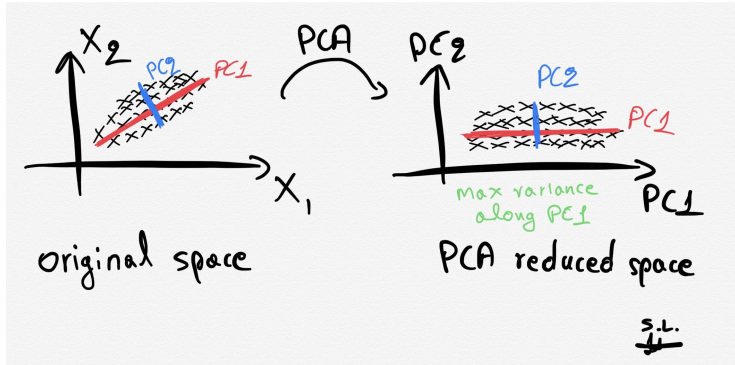
PCA as a Linear Autoencoder⁵

- Linear network $\hat{\mathbf{x}} = \mathbf{UV}\mathbf{x}$: encoder $\mathbf{V} \in \mathbb{R}^{M \times p}$, decoder $\mathbf{U} \in \mathbb{R}^{p \times M}$, $M \ll p$.
- Train on centred inputs to minimise $\frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i - \mathbf{UV}\mathbf{x}_i\|^2$.
- Take $\mathbf{U} = \mathbf{\Phi}$, $\mathbf{V} = \mathbf{\Phi}^\top$ with $\mathbf{\Phi} = [\phi_1, \dots, \phi_M]$ an orthonormal basis of an M -dimensional subspace.
- The reconstruction error is minimised exactly when $\phi_1, \dots, \phi_M$ are the first M principal components: minimal reconstruction error $\Leftrightarrow$ maximal retained variance.
- So the encoder is PCA: $\mathbf{z}_i = \mathbf{\Phi}^\top \mathbf{x}_i$ are the scores; the columns of $\mathbf{\Phi}$ are the loadings.
- Without the orthonormality constraint, $\mathbf{UV}$ still projects onto the same subspace, but the code axes are not unique.



⁵https://www.cs.toronto.edu/~rgrosse/courses/csc321_2017/slides/lec20.pdf

Summary of PCA



- What are the differences between PCA and feature selection/regularisation?
- Why does PCA belong to unsupervised learning?

Activity 1



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Watch the demonstration of the PCA, and apply the PCA to the scaled dataset of digits.

Outline



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- 1 The Curse of Dimensionality
- 2 Principal Component Analysis
- 3 T-SNE**
- 4 Principal Component Regression

When PCA does not Work

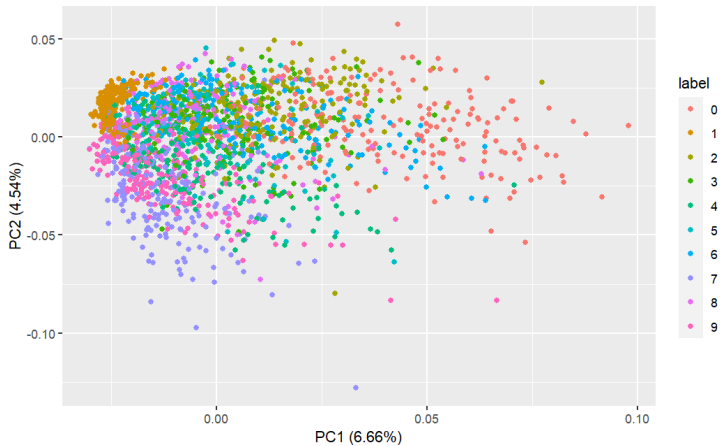


Figure: Visualisation with PCA on a sample of MNIST

- t-SNE (t-distributed Stochastic Neighbor Embedding) is an unsupervised non-linear dimensionality reduction technique
- Key ideas: preserving the relationships between data points in a lower-dimensional space
- Algorithm sketch
 - ▶ Model a point being selected as a neighbour of another point in both higher and lower dimensions. It starts by calculating a pairwise similarity between all data points in the high-dimensional space using a Gaussian kernel. The points that are far apart have a lower probability of being picked than the points that are close together. The pairwise similarity in the low dimensional space is computed by a t-distribution kernel.
 - ▶ Map higher dimensional data points into lower dimensional space while preserving the pairwise similarities. It is achieved by minimizing the divergence between the probability distribution of the original high-dimensional and lower-dimensional with gradient descent. The outcomes are the lower-dimensional embeddings

More on t-SNE (Optional)⁶

Given n observations

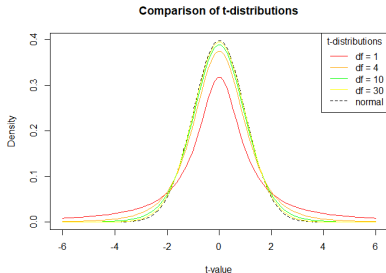
- High dimensional space:

- ▶ Define $p_{j|i} = \frac{\exp(-\|x_i - x_j\|^2 / 2\tau_i^2)}{\sum_{k \neq i} \exp(-\|x_i - x_k\|^2 / 2\tau_i^2)}$

- ▶ Define a symmetric conditional probability $p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n}$

- Low dimensional space: define $q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq i} (1 + \|y_i - y_k\|^2)^{-1}}$ for $i \neq j$

- Minimise $\sum_i KL(P_i \| Q_i) = \sum_{n=1}^n \sum_{j=1}^n p_{ij} \log \frac{p_{ij}}{q_{ij}}$



⁶Lecture notes on t-SNE.

T-SNE VS. PCA

- PCA

- ▶ is a linear dimensionality reduction technique
- ▶ tries to preserve the global structure of the data
- ▶ is highly affected by outliers
- ▶ can be used for feature extraction and visualisation

- t-SNE

- ▶ is a non-linear dimensionality reduction technique
- ▶ tries to preserve the local structure of data
- ▶ can handle outliers
- ▶ Need to tune multiple hyperparameters
- ▶ is mainly used for visualisation and exploratory data analysis

Python Demo.

Outline



MONASH University

- 1 The Curse of Dimensionality
- 2 Principal Component Analysis
- 3 T-SNE
- 4 **Principal Component Regression**

Use PCs in linear regression

- Basic idea: transform the predictors and then fit a least squares model using the transformed variables.
 - Let $Z_1, Z_2, \dots, Z_m$ represent $M < p$ linear combinations of our original p input variables (predictors).

$$Z_m = \sum_{j=1}^p \phi_{j,m} X_j = \boldsymbol{\phi}_m^T \mathbf{X}$$

where $\boldsymbol{\phi}$ are some constants to be learned.

- Then, the linear model to be fit is

$$y_i = \theta_0 + \sum_{m=1}^M \theta_m z_{i,m} + \epsilon_i, \quad i = 1, \dots, n,$$

where

$$z_{i,m} = \boldsymbol{\phi}_m^T \mathbf{x}_i$$

- The regression coefficient: $\theta_1, \theta_2, \dots, \theta_M$.

Use PCs in linear regression — continued

- Dimension reduction serves to constrain the estimated coefficients, β_j .

$$\begin{aligned}
 \sum_{m=1}^M \theta_m z_{i,m} &= \sum_{m=1}^M \theta_m \sum_{j=1}^p \phi_{j,m} x_{i,j} \\
 &= \sum_{m=1}^M \sum_{j=1}^p \theta_m \phi_{j,m} x_{i,j} \\
 &= \sum_{j=1}^p \sum_{m=1}^M \theta_m \phi_{j,m} x_{i,j} \\
 &= \sum_{j=1}^p \beta_j x_{i,j}
 \end{aligned}$$

- Can win in the bias-variance tradeoff
 - ▶ If $p \gg n$, $m \ll p$ significantly reduce the variance of the estimated coefficients.
 - ▶ if $p = n$ and all Z_m are linearly independent, the model with dimension reduction is equivalent to the least square model.

Principal Component Regression

- Basic idea:

- 1 Construct the first M principal components, $Z_1, Z_2, \dots, Z_M$
- 2 Fit a linear regression with the M components as predictors

$$Y = \beta_0 + \sum_{m=1}^M \beta_m Z_m + \epsilon$$

- Assumption: the directions in which $X_1, X_2, \dots, X_p$ show the most variation are the directions that are associated with Y .
 - ▶ Not guaranteed to be true, but a reasonable enough approximation
 - ▶ If the assumption holds, fitting a least square model to the M principal components is better than to the P variables.
 - ▶ Mitigate overfitting: $M \ll P$

Principal Component Regression: Simulated data

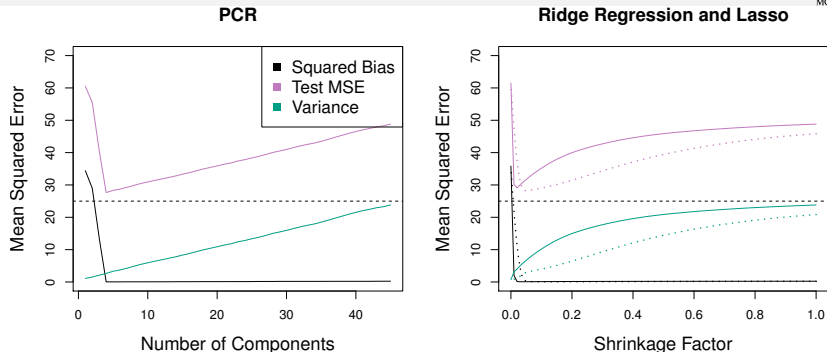


Figure: Left: results for PCR. Right: Results for lasso (solid) and ridge regression (dotted)

- A simulated dataset that is favourable to PCR.
 - ▶ The response variable is a function of the first five principal component.
- The plots show
 - ▶ The bias drops to zero rapidly as M increases.
 - ▶ $M = 5$ gives the minimum MSE.
- **Question:** Why PCR sometimes does not perform as well as Ridge and Lasso?

Principal Component Regression: $M=?$

- PCR is not a feature selection method.
 - ▶ A linear combination of all p of the original features (or predictors)
 - ▶ Closely related to ridge regression than to the lasso
- How many principal components should we use?

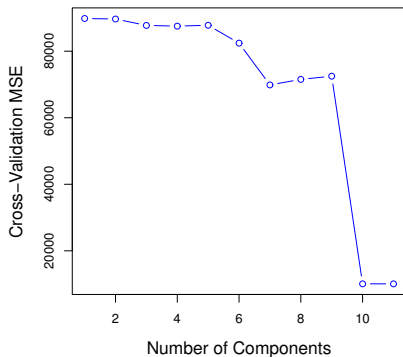


Figure: The 10-fold cross validation MSE obtained using PCR, as a function of M .

Principal Component Regression: Standardising Features

- The scale on which the variables are measured ultimately has an effect on the final PCR model
 - ▶ measures of tree size
 - the trunk diameter in cm, biomass of leaves in kg, number of branches, overall height in meters
- Variables with the highest sample variances will tend to be emphasised in the first few principal components.
- Principal component analysis using the covariance function should only be considered if all of the variables have the same units of measurement.
- Each component's eigenvalue represents how much variance it explains.
- Standardized variables before PCA:

$$\hat{X}_{i,j} = \frac{X_{i,j} - \bar{x}_j}{s_j}$$

where s_j is the standard deviation of the j -th variable.



Summary

- What goes wrong in high dimensions?
- When do we need dimension reduction?
- PCA
 - ▶ What are the key ideas behind PCA?
 - ▶ It captures linear relationships between principal component scores and input variables (predictors)
- t-SNE
 - ▶ What are the key ideas behind t-SNE?
 - ▶ It captures non-linear relationships between principal component scores and input variables (predictors)
- PCR
 - ▶ What are the main difference between linear regression and PCR?
 - ▶ What are its strengths and weaknesses?

Summary

- "Linear Model Selection and Regularisation", Chapter 6 of "Introduction to Statistical Learning"
 - ▶ Section 6.3 "Dimension Reduction Methods"
 - ▶ Section 6.4 "Considerations in High Dimensions"
- "Unsupervised learning", Chapter 12 of "Introduction to Statistical Learning, with Application in R/Python"
 - ▶ Section 12.2 "Principal Components Analysis"

Acknowledgement:

- Figures in this presentation were taken from "An Introduction to Statistical Learning, with applications in R/Python" (Springer, 2023) with permission from the authors: G. James, D. Witten, T. Hastie and R. Tibshirani
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